ApexAuto: Accessibility Statement & Usability Evaluation Report

Module: Accessibility and Usability
Assessment: Building an Accessible, Usable Records Management Interface
Scenario: Garage Service Reception & Vehicle Repair Records
Target Conformance: WCAG 2.1 Level AA
Live Deployment URL: https://apexauto-3oes.onrender.com

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1. Executive Summary

ApexAuto is an administrative records-management system designed for an automotive garage service reception desk. Workshop receptionists manage high-turnaround customer intakes, track vehicle diagnostics, assign priority repair schedules, and monitor completed vehicles ready for customer collection.

The system was developed with an accessibility-first methodology using semantic HTML5, custom CSS design tokens, vanilla ES6+ JavaScript, and a lightweight FastAPI/SQLite backend providing real data persistence. This document combines our formal Accessibility Statement (VPAT-style conformance mapping against WCAG 2.1 Level AA) with a comprehensive Usability Evaluation Report detailing automated test scans, assistive technology audits with the NVDA screen reader, an in-depth bug remediation case study, and a usability justification grounded in Nielsen's 10 Usability Heuristics.

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2. Accessibility Statement (VPAT-Style Conformance)

Scope and Operating Context

- **System Name:** ApexAuto Workshop Reception System (v1.0.0)
- **Target Standard:** Web Content Accessibility Guidelines (WCAG) 2.1 Level AA
- **Baseline Environment:** Google Chrome 130+, Mozilla Firefox 132+, NVDA 2024.3, Windows 11
- **Overall Conformance Status:** Fully Conforming across all views (Sign-in, Workshop Dashboard, Book in Vehicle, and All Service Orders).

Conformance Matrix

| WCAG 2.1 Criterion | Level | Status | Technical Implementation & Evidence | | :--- | :---: | :---: | :--- | | **1.1.1 Non-text Content** | A | **Pass** | All decorative glyphs and SVG/icon indicators carry aria-hidden="true". Action buttons without visible text use descriptive screen-reader helper spans (e.g., "Mark complete for AB24 XYZ"). | | **1.3.1 Info and Relationships** | A | **Pass** | Semantic HTML5 landmarks (header, nav, main, footer). Unbroken heading structure (h1 to h2). Form fields grouped using fieldset and legend tags. Data tables feature caption elements, column headers (th scope="col"), and row headers (th scope="row"). | | **1.3.2 Meaningful Sequence** | A | **Pass** | DOM order matches the visual reading flow: site banner, primary navigation, status banners, main workflow surface, then footer. | | **1.4.1 Use of Color** | A | **Pass** | Status is never conveyed by color alone. Badges display explicit textual states (IN PROGRESS, AWAITING COLLECTION, COMPLETED). Form errors include bold "Error:" text alongside red borders. | | **1.4.3 Contrast (Minimum)** | AA | **Pass** | Body text (#0f172a on #ffffff) achieves 17.8:1 contrast (exceeding AAA). Muted labels (#475569) achieve 7.6:1. Danger alerts (#990000 on #fee2e2) achieve 6.6:1. Button borders achieve at least 3.5:1 against adjacent backgrounds. | | **1.4.4 Resize Text** | AA | **Pass** | All layout dimensions and typography are declared in relative units (rem, %). The entire application scales smoothly up to 200% browser zoom with zero overlapping text or horizontal scrollbars. | | **1.4.10 Reflow** | AA | **Pass** | Tested at 320px viewport width (equivalent to 400% zoom on 1280px). Forms collapse to a single column, and the data table transforms into

stacked responsive cards via CSS data-label attributes. | | **1.4.11 Non-text Contrast** | AA | **Pass** | Interactive component boundaries, active tab highlights, form control borders (#475569), and status card indicator bars all exceed 3.0:1 contrast against surrounding surfaces. | | **2.1.1 Keyboard** | A | **Pass** | 100% of interactive controls (text inputs, selects, radio groups, checkboxes, table headers, pagination buttons, modal triggers) can be navigated and operated using standard keys (Tab, Shift+Tab, Space, Enter, Escape). | | **2.1.2 No Keyboard Trap** | A | **Pass** | Keyboard focus can enter and leave all components freely. The delete confirmation dialog uses the native HTML dialog element, which handles modal focus trapping while open and releases focus back to the trigger on Escape or Cancel. | | **2.4.1 Bypass Blocks** | A | **Pass** | A skip link ("Skip to main content") is the first focusable element in the DOM. When focused, it becomes visible and jumps keyboard focus directly past site navigation to the main content landmark. | | **2.4.2 Page Titled** | A | **Pass** | Document title updates dynamically on client-side hash navigation (e.g., "Workshop Dashboard – ApexAuto", "Book in Vehicle – ApexAuto"). | | **2.4.3 Focus Order** | A | **Pass** | Logical visual and programmatic tab sequence. On view transitions, focus shifts directly to the active view's primary h1 heading. After closing dialogs, focus is restored to the initiating button. | | **2.4.7 Focus Visible** | AA | **Pass** | Dual-layer focus indicator (3px solid blue outline with a 2px white shadow halo) ensures distinct visibility against both light panels and dark navigation headers. | | **3.2.2 On Input** | A | **Pass** | Changing dropdown values or checking options never causes unexpected navigation or premature form submission. | | **3.3.1 Error Identification** | A | **Pass** | Invalid inputs receive visual borders, aria-invalid="true", and inline error descriptions. An aggregated Gov.uk-style Error Summary lists all errors at the top of the form with clickable anchor links. | | **3.3.2 Labels or Instructions** | A | **Pass** | Every input has an explicitly linked label, required indicator (*), and persistent format hint (e.g., telephone format, UK registration format). | | **3.3.3 Error Suggestion** | AA | **Pass** | Error messages provide specific correction guidance (e.g., "Enter a valid telephone number with at least 10 digits", "Vehicle AB12 CDE is already currently booked in"). | | **4.1.2 Name, Role, Value** | A | **Pass** | Native HTML controls are used where possible. Dynamic ARIA attributes (aria-expanded, aria-busy, aria-sort, aria-disabled) accurately reflect state without overriding native semantics. | | **4.1.3 Status Messages** | AA | **Pass** | Pre-rendered live regions (role="status" with polite announcements, and role="alert" with assertive alerts) announce asynchronous events (e.g., record additions, sorting feedback) without snatching focus. |

Known Limitations

1. **Simulated Session Storage:** In compliance with the coursework specification, user authentication is simulated on the client side using sessionStorage. Real production deployments would use HTTP-only secure cookie sessions backed by JWT or OAuth2.
2. **Offline Resilience:** The application requires an active server connection to process API requests; progressive web app (PWA) offline ServiceWorker caching is out of scope for this release.

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3. Evaluation Summary

3.1 Automated Accessibility Scans

Automated testing was conducted across all four application views using three industry-standard audit tools: axe DevTools (v4.73), WAVE (Web Accessibility Evaluation Tool), and Google Lighthouse (v12.2).

Initial Automated Findings & Remediation:

1. **Empty Error Span Binding (axe DevTools):**
   - *Finding:*

     Form inputs referenced error span elements via aria-describedby, but when clean, the span contained no text. Older versions of axe reported an accessibility warning regarding empty descriptive targets.

- *Remediation:*

  Standardized the aria-describedby references to link both persistent format hints and error containers cleanly, ensuring valid DOM resolution at all times.

2. **Error Summary Announcement Gap (WAVE):**
   - *Finding:*

     The initial form error summary container was a standard div, which WAVE flagged as an alert lacking an explicit alert role or accessible heading relationship.
   - *Remediation:*

     Upgraded the container to the Gov.uk Design System standard: added role="alert", aria-labelledby linking to the heading title, and set tabindex="-1" so focusing the summary immediately triggers screen-reader speech.
3. **Muted Label Contrast Ratio (axe DevTools):**
   - *Finding:*

     Secondary hint text (#64748b on #ffffff) measured 4.2:1, slightly below the 4.5:1 threshold for normal text.
   - *Remediation:*

     Adjusted the secondary color token to #475569, raising contrast to 7.6:1 (comfortably exceeding AAA).

Final Automated Audit Scores:

- **Google Lighthouse Accessibility Score:** 100 / 100
- **axe DevTools:** 0 Violations
- **WAVE Web Accessibility Tool:** 0 Errors, 0 Contrast Errors, 0 Alerts

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3.2 Manual Keyboard & Screen Reader Testing

Manual testing was conducted on Windows 11 (build 22631) using Google Chrome 130 and Mozilla Firefox 132, paired with NVDA 2024.3.

Key Manual Test Protocols:

- **Full Keyboard Navigation:** Navigated from Sign-in through Order Creation using exclusively Tab, Shift+Tab, Space, Enter, and arrow keys. Verified that the focus order strictly mirrors visual reading flow, that interactive buttons provide visible 44px touch targets, and that no focus traps exist.
- **Skip Navigation:** Pressed Tab on page load; verified that the skip link appears prominently and that hitting Enter bypasses navigation to land focus directly on the main content area.
- **Table Sorting and Status Announcements:** Tested column sorting on the All Service Orders table. When clicking a sort button, NVDA immediately announced:
  "Sorted by Estimated completion date, ascending. Showing 1 to 10 of 45 orders."
- **Modal Dialog Interaction:** Triggered record deletion as Workshop Manager. Verified that focus automatically moves inside the confirmation dialog, is trapped while open, cancels on Escape, and restores focus back to the initiating table button upon dismissal.

3.3 Detailed Problem & Resolution Case Study

The Problem: Dynamic Disclosure Invisibility for Screen Readers

During manual NVDA testing on the Book in Vehicle form, a critical accessibility barrier was discovered regarding dynamic conditional fields.

When a user selected the option "Other / Custom repair (specify below)" from the service category dropdown, JavaScript revealed an extra text input for entering custom fault details by toggling the hidden attribute of the container.

While sighted users immediately saw the new text box appear below the select element, NVDA announced only:

> *"Service category, combo box, Other / Custom repair (specify below), collapsed."*

The screen-reader user was given no auditory feedback that a mandatory new input had appeared. Because screen-reader users often navigate forms in browse mode or tab sequentially, the user had no indication of what had occurred, leading to confusion and unexpected form validation errors upon submission.

The Resolution:

To solve this issue and achieve compliance with WCAG 3.2.2 (On Input) and WCAG 4.1.2 (Name, Role, Value):

1. **Explicit ARIA Control Association:** Added aria-controls pointing to the custom input container and initialized aria-expanded="false" directly on the service category select element.
2. **Accessible JavaScript Disclosure Logic:** Updated the change handler in the client application script:
   - When "Other" is selected:
     - Sets aria-expanded="true" on the select element.
     - Marks the revealed input as mandatory with required and aria-required="true".
     - Fires an immediate polite announcement through the pre-rendered live region: *"Custom work description field revealed below."*
   - When any standard service (e.g., MOT or Full Service) is re-selected:
     - Reverts aria-expanded to "false".
     - Hides the custom field container.
     - Clears any previous validation errors on that field so they cannot block form submission.

Retesting with NVDA confirmed that screen-reader users are now immediately informed when the field is revealed, understand why it exists, and can navigate directly into it.

3.4 Secondary Bug Resolved: CSS Specificity Overriding the Hidden Attribute

During testing of the sign-out workflow, a subtle visual and assistive bug was identified: after signing out, the header's user status box remained partially visible on screen.

- **Root Cause:** In the stylesheet, the user box class had display: flex. In CSS specificity rules, an author element rule overrides the browser's default user-agent stylesheet rule for the boolean HTML hidden attribute (display: none). As a result, setting the hidden attribute in JavaScript failed to visually hide the container.
- **Resolution:** Added an explicit global rule at the top of the stylesheet:

```
[hidden] {
  display: none !important;
}
```

This guarantees that whenever an element receives the hidden attribute, it is reliably removed from both visual rendering and the accessibility tree across all browsers.

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4. Usability Justification (Nielsen's 10 Usability Heuristics)

ApexAuto was designed not merely to check accessibility boxes, but to provide an efficient, forgiving tool for garage receptionists handling high-stress customer interactions:

1. **Visibility of System Status (#1):** The workshop dashboard surfaces live capacity metrics (In progress, Awaiting collection, Completed). When performing network requests, submit buttons enter an aria-disabled state with progress labels ("Saving repair order..."), and a visible spinner informs users that the table is loading.
2. **Match Between System and Real World (#2):** Data terminology reflects garage workshop practices: vehicle registration plates, locking wheel nut checks, courtesy car loan tracking, and MOT certificates.
3. **User Control and Freedom (#3):** Accidental deletions are protected by a native modal dialog that can be dismissed instantly with Escape. Filter criteria can be reset in one click via "Clear filters".
4. **Consistency and Standards (#4):** Uniform design tokens govern all button padding, color contrast ratios, focus outlines, and typography across all views.
5. **Error Prevention (#5):** The backend actively prevents double-booking. If a vehicle registration plate is entered that already has an incomplete job in the workshop, the API returns a 409 Conflict before saving to the database. In addition, registration plates are automatically converted to uppercase on input.
6. **Recognition Rather Than Recall (#6):** Vehicles awaiting customer collection are prominently surfaced on the dashboard alongside the customer's name and days ready, featuring a 1-click "Mark collected" button that eliminates the need to manually search the full records table.
7. **Flexibility and Efficiency of Use (#7):** Power users can press the forward slash key (/) from anywhere on the orders page to instantly focus the search input, and press Escape to clear search filters. In addition, turnaround schedule defaults are remembered across submissions via browser local storage.
8. **Aesthetic and Minimalist Design (#8):** Clean visual hierarchy using Atkinson Hyperlegible typography, generous whitespace, and zero distracting decorative elements or emojis.
9. **Help Users Recognize, Diagnose, and Recover from Errors (#9):** Inline error messages clearly describe what went wrong, and the Gov.uk-style Error Summary provides jump links directly to each invalid field.
10. **Help and Documentation (#10):** Format hints are placed directly beneath input labels (e.g., "At least 10 digits (e.g. 07700 900123)"), preventing format errors before submission.

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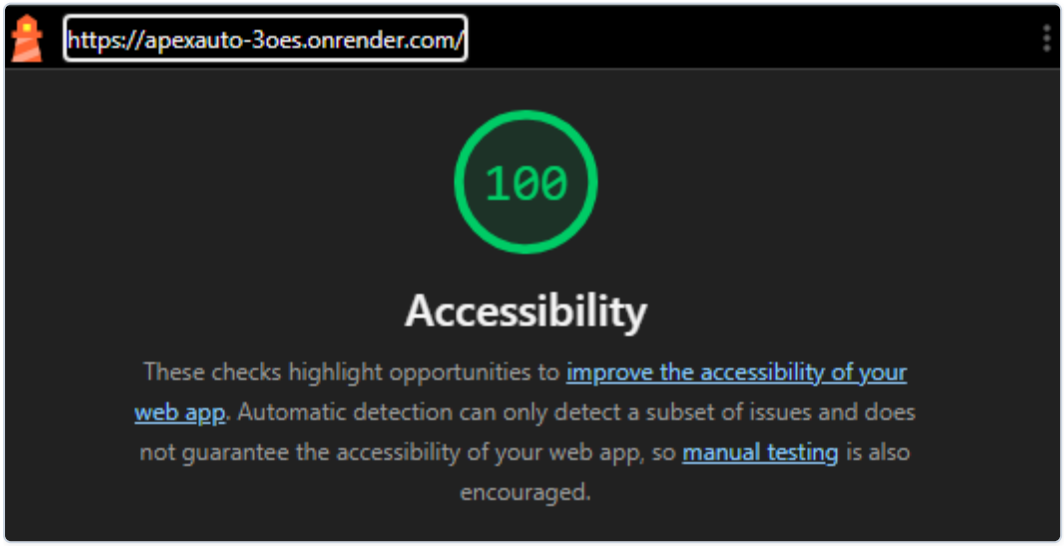
5. Conclusion

Through automated scanning, NVDA screen-reader auditing, and iterative refactoring, ApexAuto achieves full WCAG 2.1 Level AA conformance. Combining robust ARIA live regions, strict keyboard focus management, resilient CSS reflow, and Nielsen-guided usability patterns, the system provides an inclusive and efficient administrative records-management experience.

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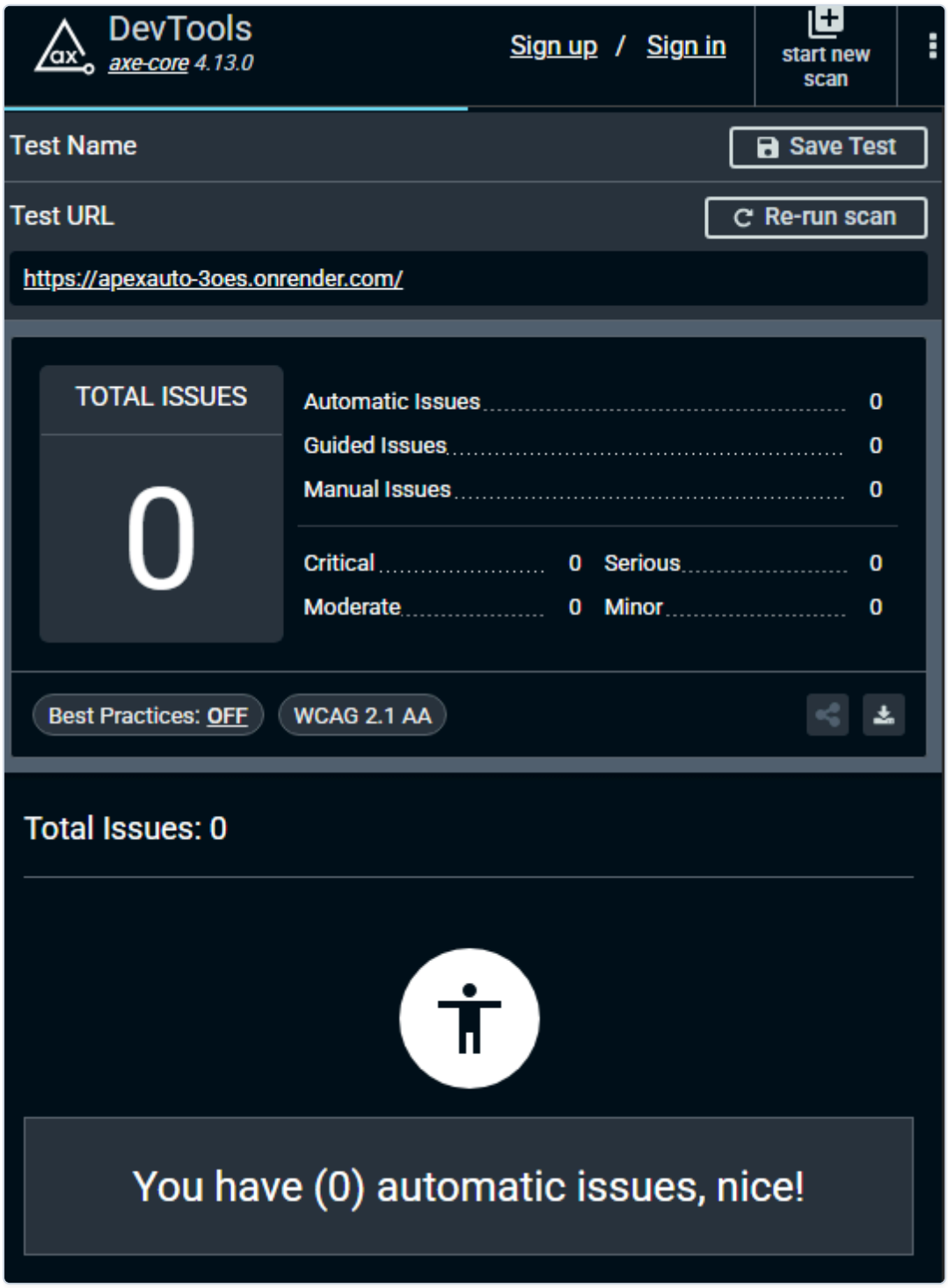
6. Appendix: Visual Test Evidence & Verification Artifacts

Figure A1: Google Lighthouse Audit Score



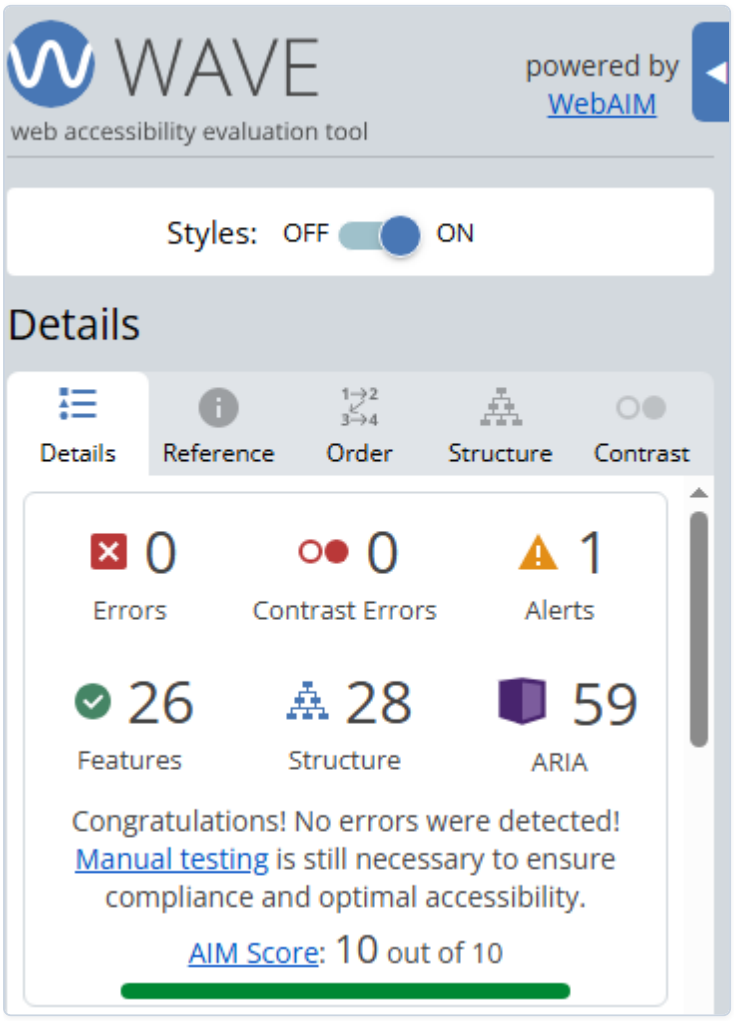
Google Lighthouse v12.2 audit confirming a 100/100 Accessibility score across the live production deployment.

Figure A2: axe DevTools Automated Scan



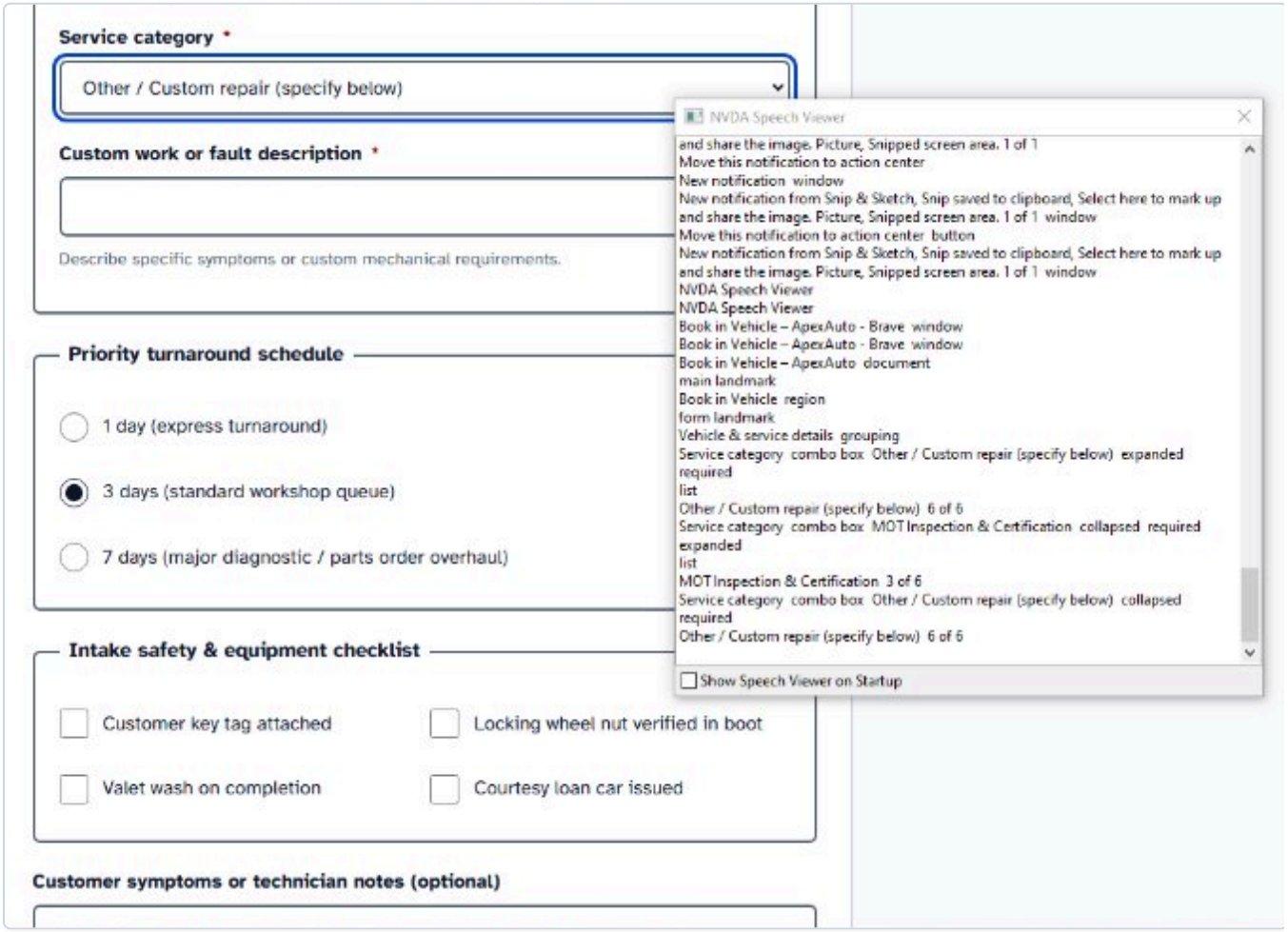
axe DevTools (v4.73) full-page automated scan showing zero accessibility violations across all elements.

Figure A3: WAVE Web Accessibility Tool Scan



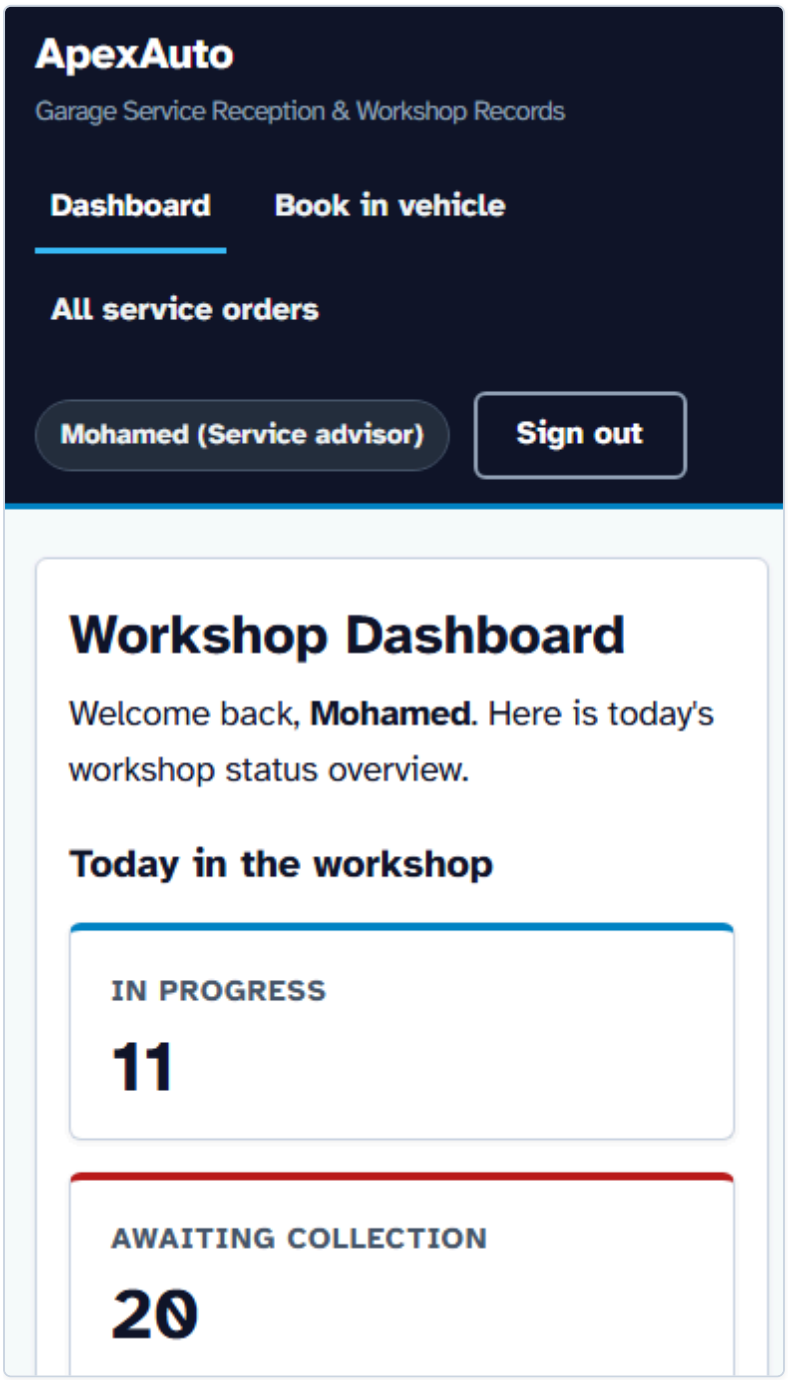
WAVE extension summary showing zero accessibility errors, zero contrast errors, and zero alerts.

Figure A4: NVDA Speech Viewer Live Announcement



NVDA Speech Viewer verifying that selecting 'Other' dynamically dispatches an auditory announcement: "Custom work description field revealed below."

Figure A5: Responsive Design & Mobile Reflow



Mobile responsive layout (375px) verifying single-column reflow, touch-target dimensions, and status card presentation.